

Project Design Phase
Proposed Solution Template

Date	18 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Small-to-medium retail enterprises face barriers with complex, bloated platforms that force rigid templates across all products. Storefronts cause high cognitive load with clunky interfaces, and shopping carts are frequently lost when users browse across devices, leading to cart abandonment. Admin tools lack simple printing or data export features.
2.	Idea / Solution description	ShopEZ is a lightweight, responsive MERN stack e-commerce platform. It features an adaptive storefront that dynamically changes layout based on product categories and maintains persistent shopping carts via local cache sync. The integrated admin portal offers real-time metrics, smart catalog forms, single-click invoice printing, and CSV exports.
3.	Novelty / Uniqueness	The platform uses context-aware category adaptation, automatically hiding irrelevant sizing selectors for electronics or decor, simplifying both the storefront and the admin inventory forms. It features isolated media print styling for clean invoices and automatic role-based guarding that routes admins and customers to their respective interfaces upon login.
4.	Social Impact / Customer Satisfaction	Lowens the barrier to digital transformation for local mom-and-pop retailers by offering a self-hosted, lightweight software suite. Optimizes the shopping experience by reducing buyer fatigue, preventing incorrect size orders, and maintaining intact carts.
5.	Business Model (Revenue Model)	A self-hosted SaaS model with a tiered subscription (free basic, paid advanced features like multi-store sync). Includes a transaction fee model retaining a 0.5% - 1.5% commission per successful payment. Additional revenue from premium customizations like custom themes or local POS sync plugins.

6.	Scalability of the Solution	Frontend (React/Vite) compiles to static assets for global CDN distribution (Vercel/Cloudflare). Backend (Node/Express) uses stateless REST APIs for Docker containerization and horizontal scaling via Load Balancers. MongoDB scales via Replica Sets and database sharding utilizing custom keys.
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